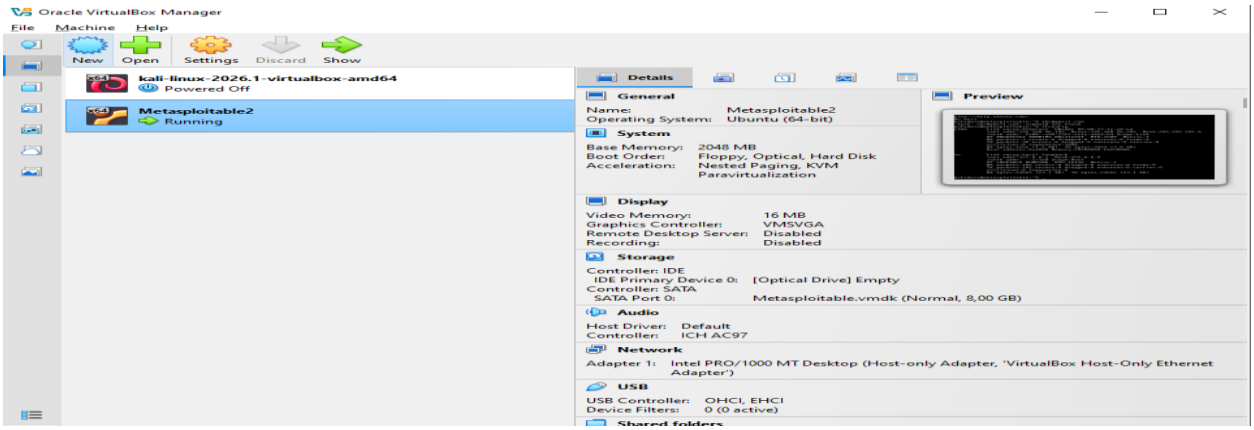
Syed Muhammad Abdullah Shah

Roll No. HM-B2-2026-1037-296

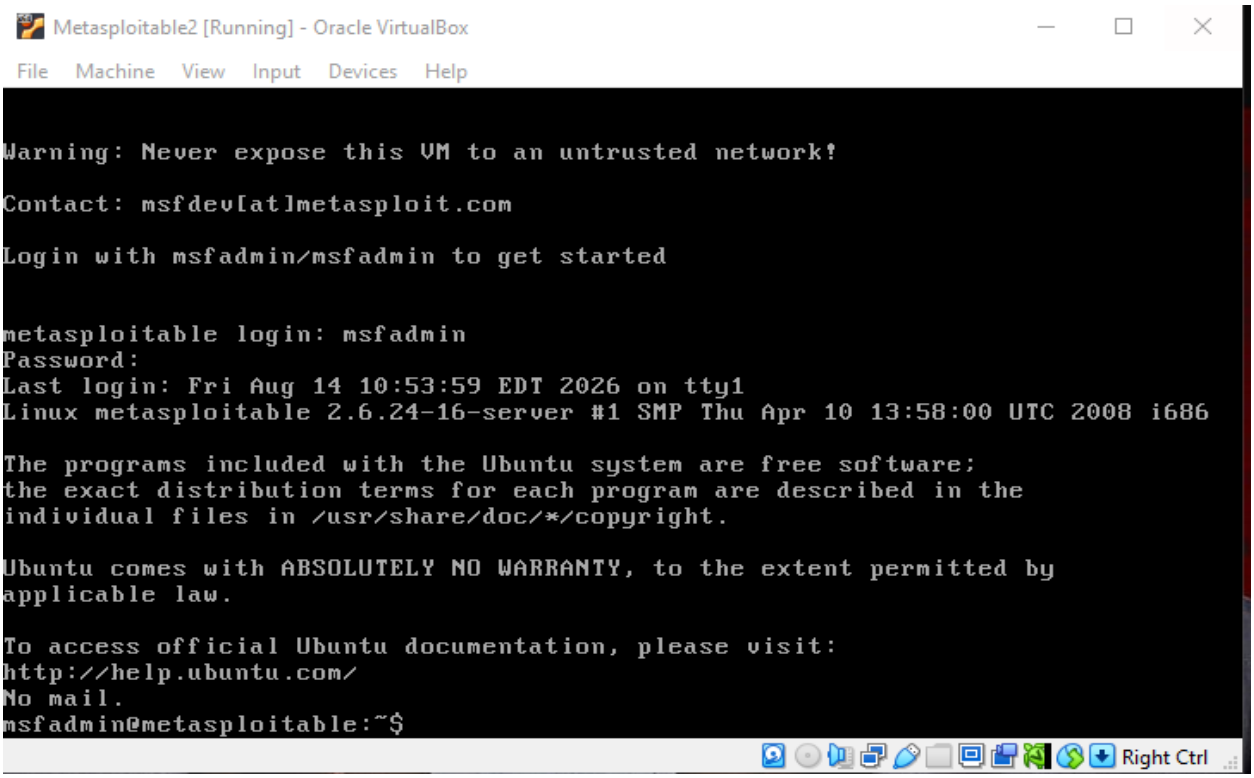
Q.1

Downloading Metasploitable2 from SourceForge as per requirements.

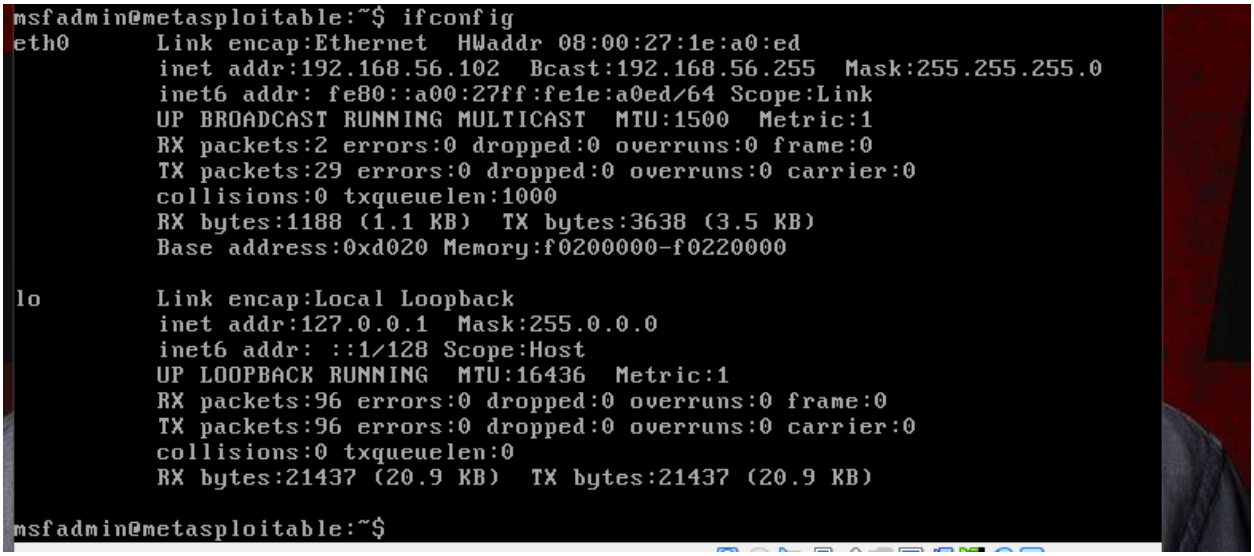
I created new and added the file using specify virtual hard disk



Afterwards, I used msfadmin as password and login.

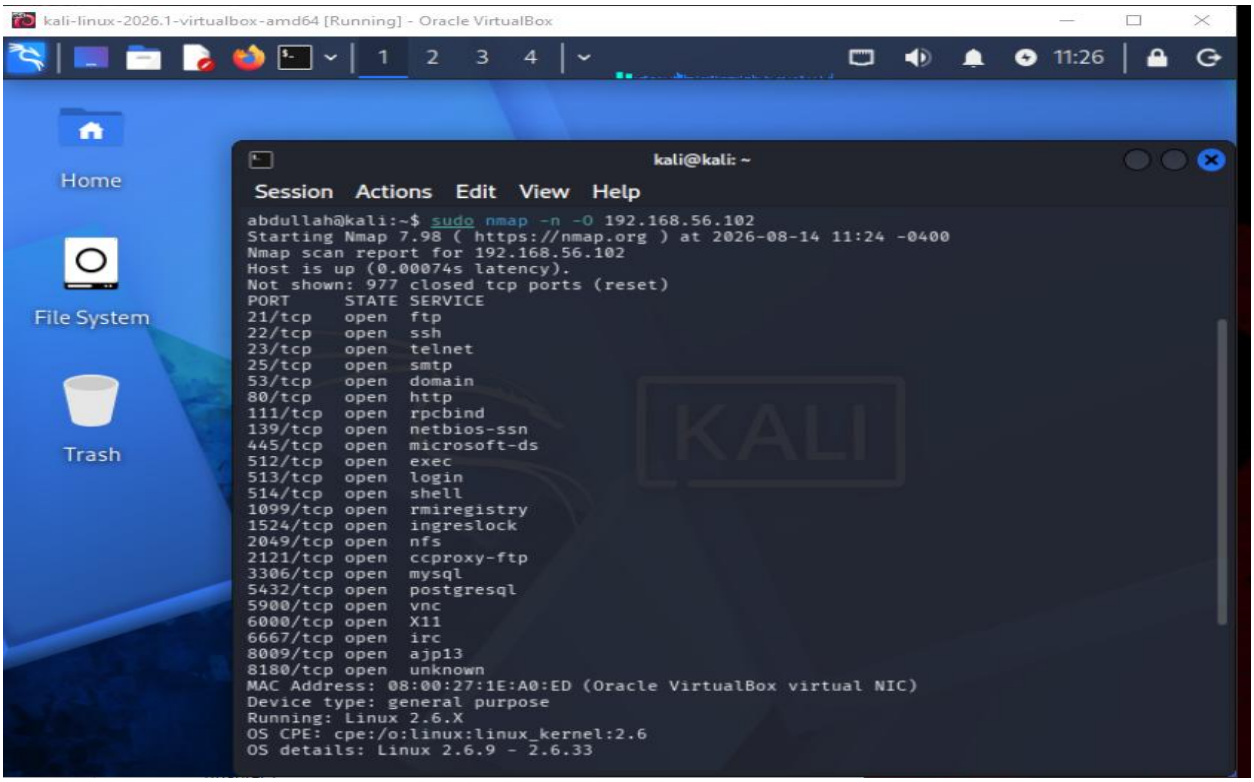


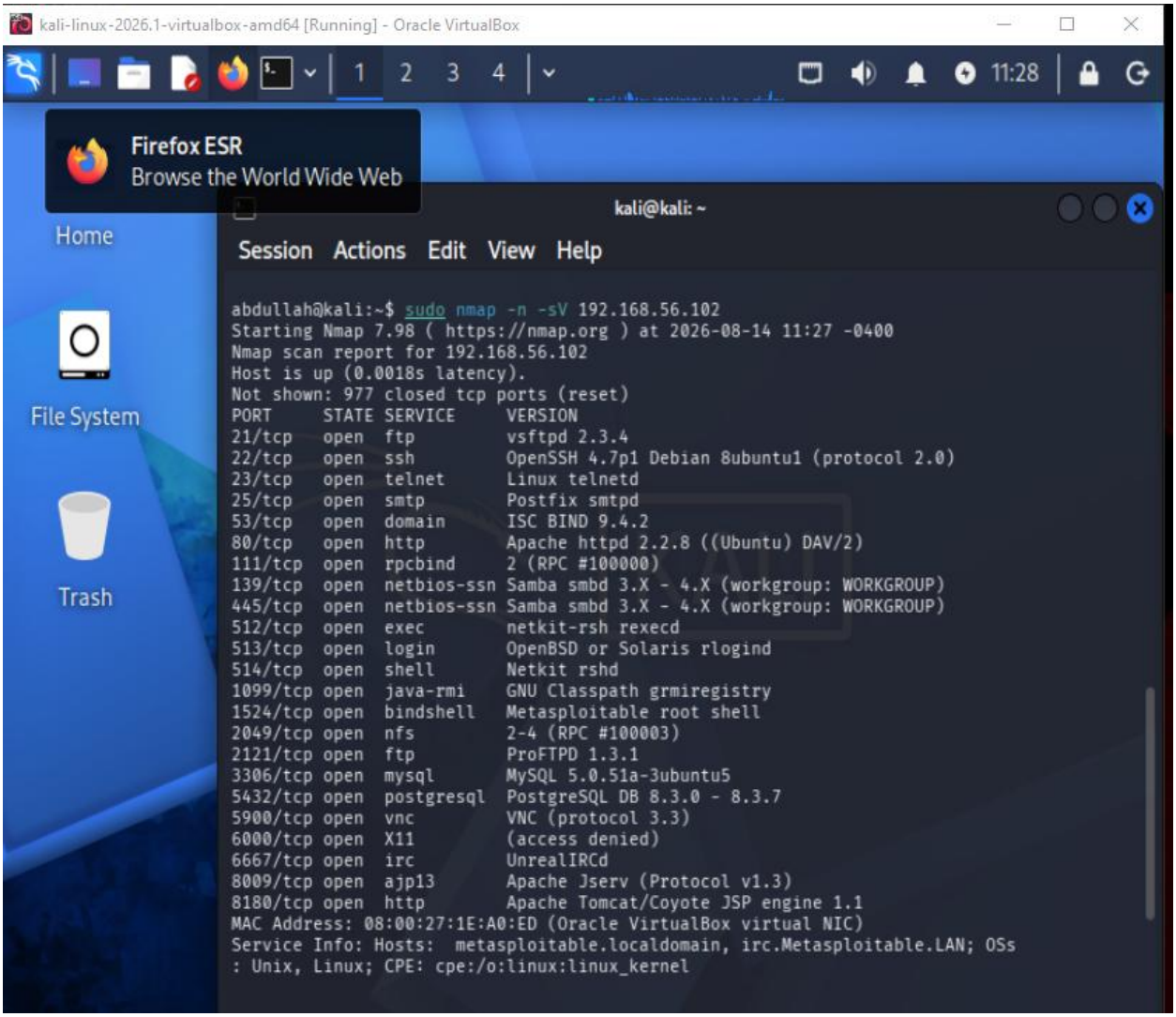
I used ifconfig to check ip 192.168.56.102 **so that I can use it as a target for my Nmap scans in Q2.**



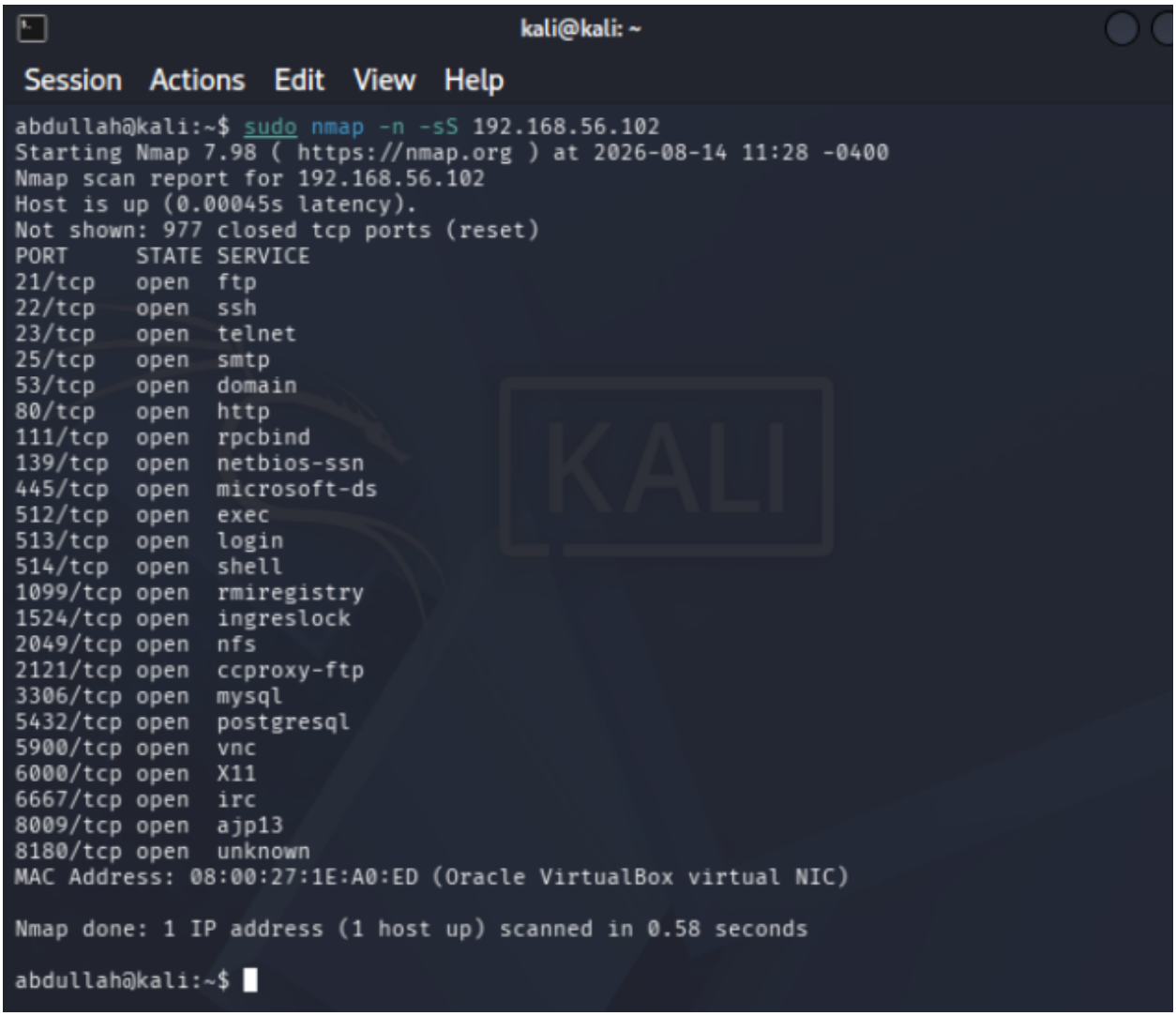
Q.2

With sudo nmap -n -O 192.168.56.102 command I can detect the Operating system of that ip which is Linux. So this way I know my target machine OS.

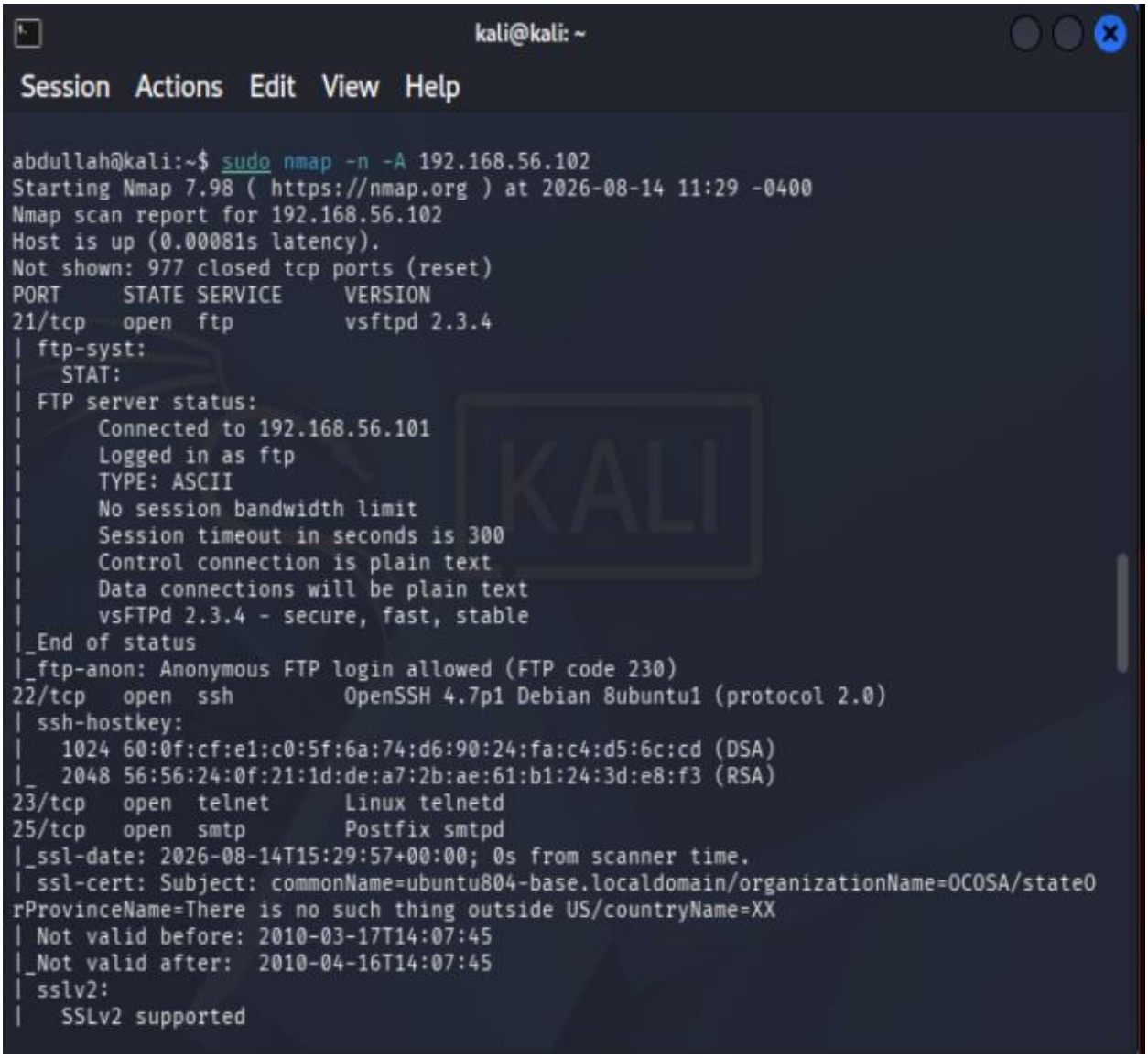




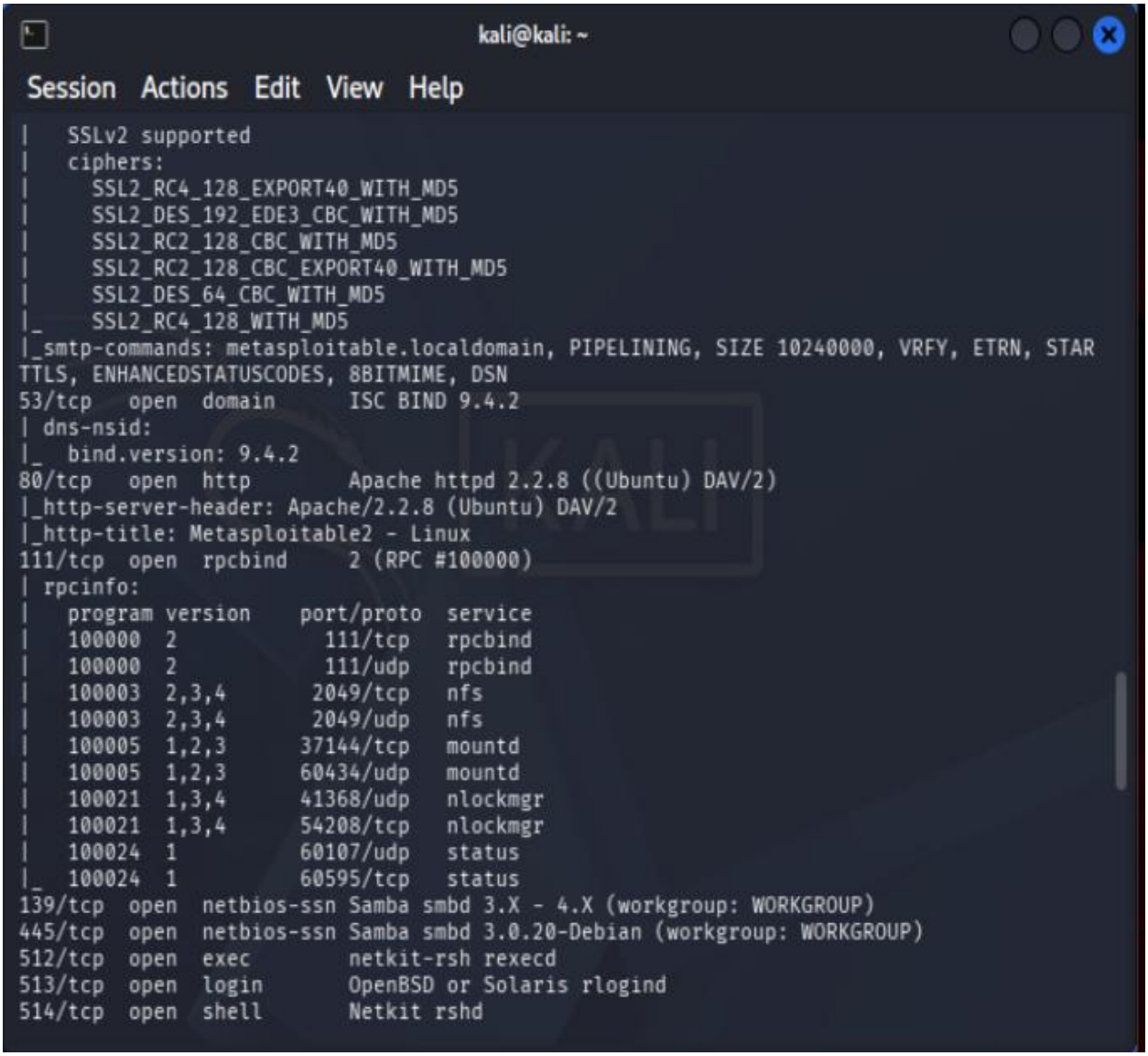
With sudo nmap -n -sV 192.168.56.102 command I can check which service version detection Scan and it help me know which version is running on that operating system of IP address which in this case is Ubuntu. That help with vulnerability identification.

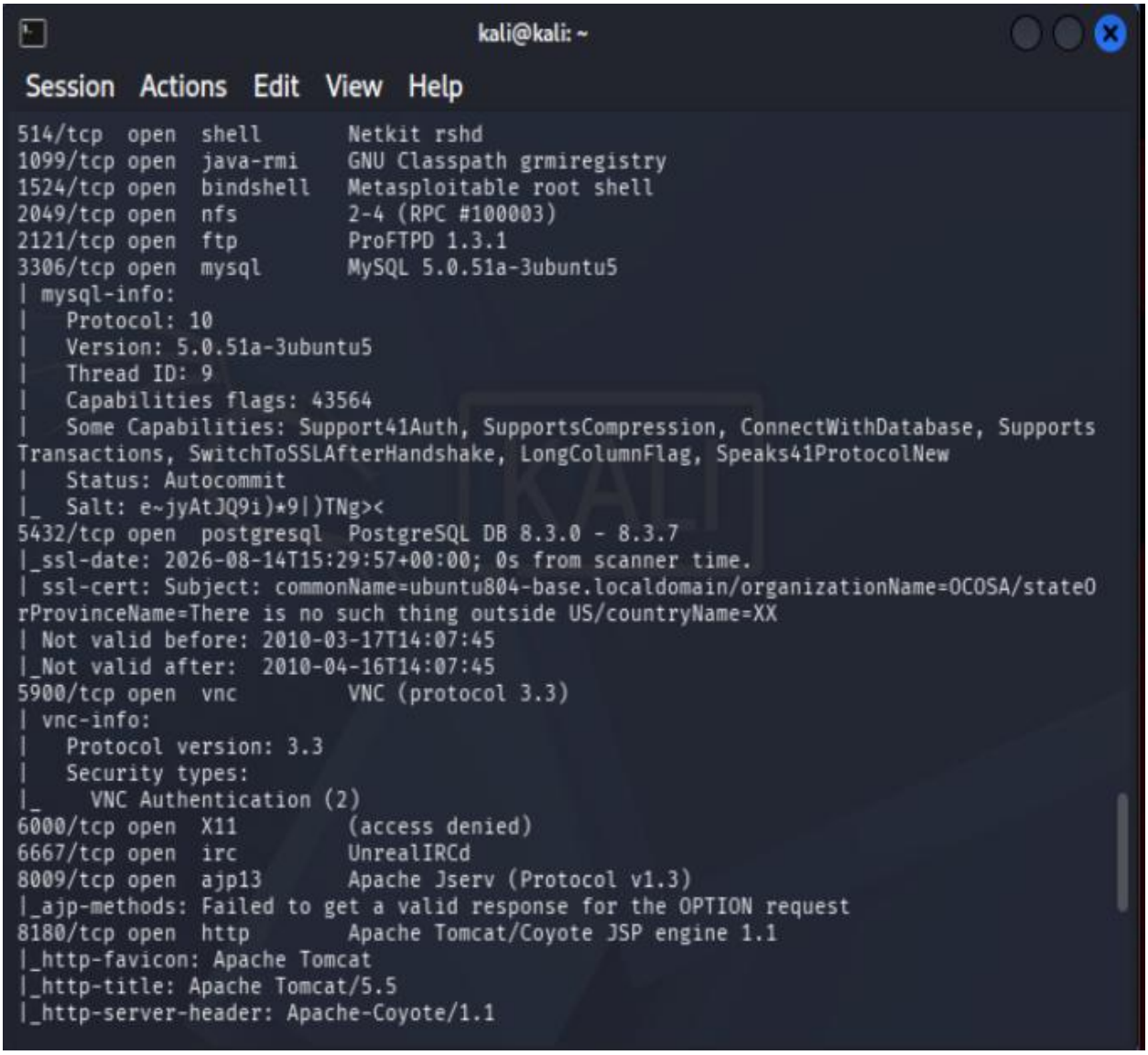


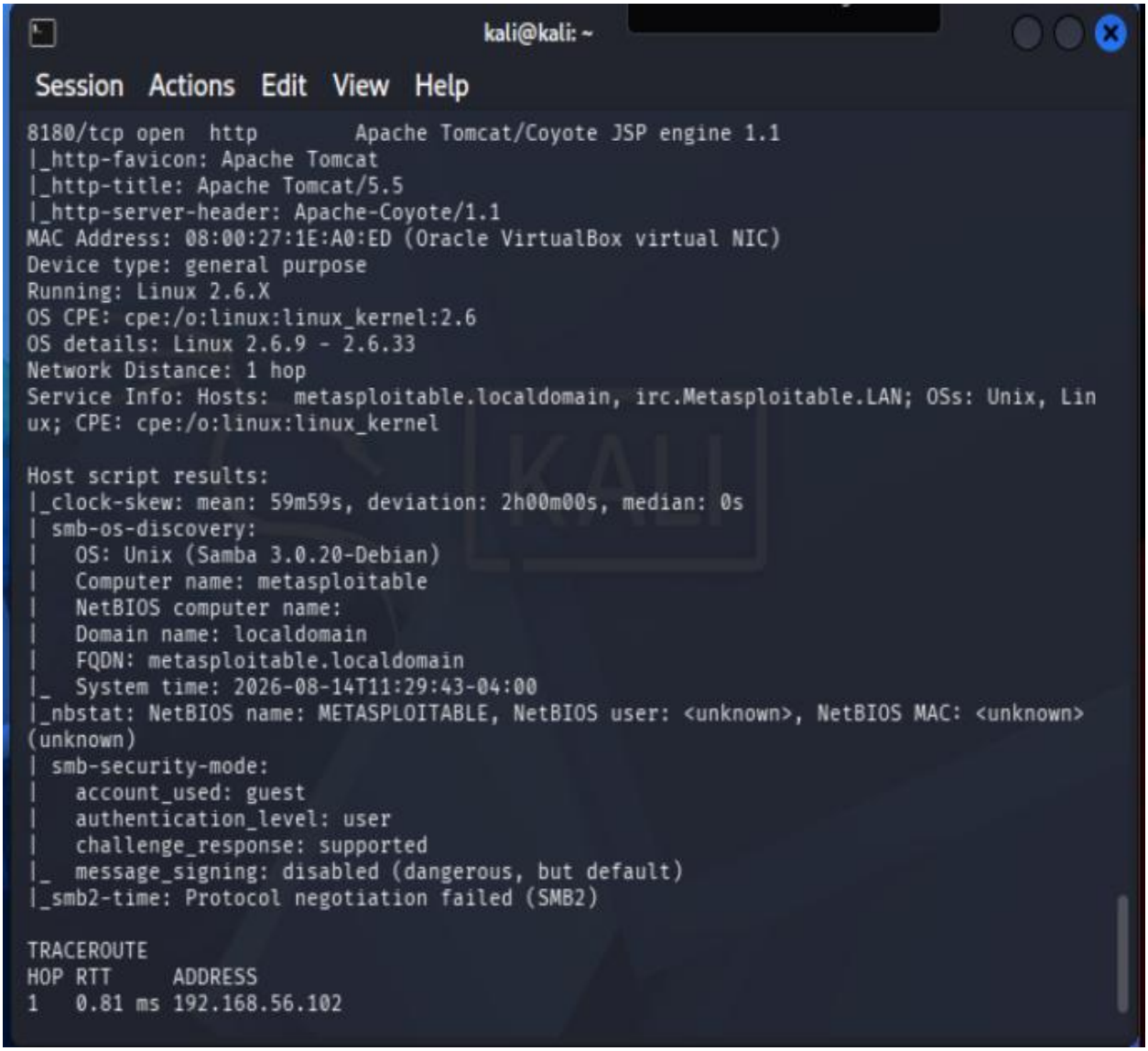
With sudo nmap -n -sS 192.168.56.102 command I can perform Main TCP SYN Stealth Scan. We can detect Target machine open TCP ports. It is fast and help to bypass simple firerwall detection.

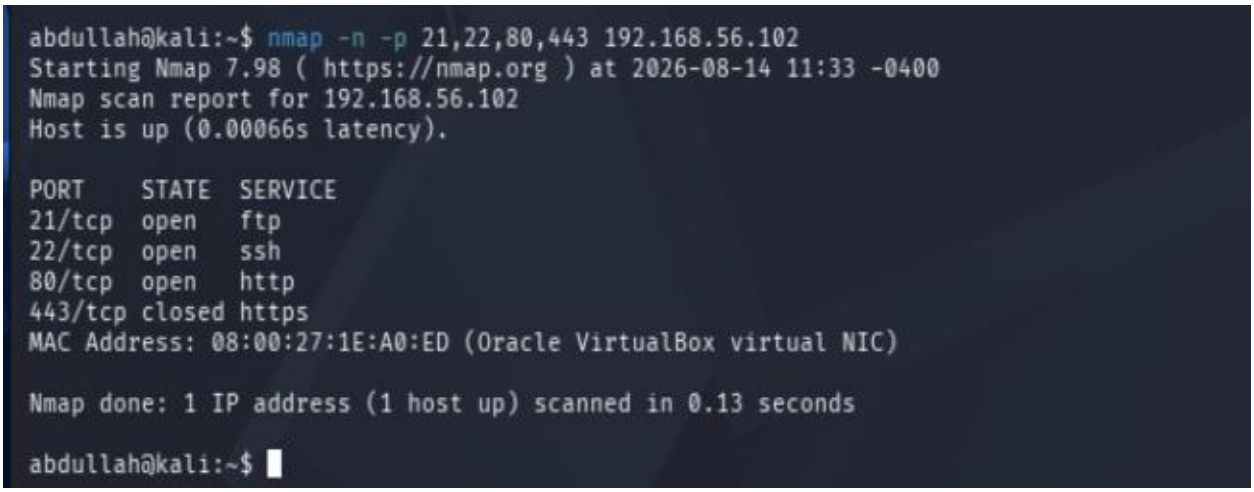


sudo nmap -n -A 192.168.56.102 help me to run Main Aggressive scan , "all-in-one" comprehensive scan that do OS detection, version detection, script scanning, and traceroute. It is detailed output


kali@kali: ~
Session Actions Edit View Help
| SSLv2 supported
| ciphers:
| SSL2_RC4_128_EXPORT40_WITH_MD5
| SSL2_DES_192_EDE3_CBC_WITH_MD5
| SSL2_RC2_128_CBC_WITH_MD5
| SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
| SSL2_DES_64_CBC_WITH_MD5
|_   SSL2_RC4_128_WITH_MD5
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp   open domain   ISC BIND 9.4.2
| dns-nsid:
|_ bind.version: 9.4.2
80/tcp   open http   Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp open rpcbind   2 (RPC #100000)
| rpcinfo:
| program version   port/proto service
| 100000 2   111/tcp   rpcbind
| 100000 2   111/udp   rpcbind
| 100003 2,3,4   2049/tcp   nfs
| 100003 2,3,4   2049/udp   nfs
| 100005 1,2,3   37144/tcp   mountd
| 100005 1,2,3   60434/udp   mountd
| 100021 1,3,4   41368/udp   nlockmgr
| 100021 1,3,4   54208/tcp   nlockmgr
| 100024 1   60107/udp   status
|_ 100024 1   60595/tcp   status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec   netkit-rsh rexecd
513/tcp open login   OpenBSD or Solaris rlogind
514/tcp open shell   Netkit rshd


kali@kali: ~
Session Actions Edit View Help
514/tcp open shell Netkit rshd
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
| mysql-info:
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 9
| Capabilities flags: 43564
| Some Capabilities: Support41Auth, SupportsCompression, ConnectWithDatabase, Supports
Transactions, SwitchToSSLAfterHandshake, LongColumnFlag, Speaks41ProtocolNew
| Status: Autocommit
|_ Salt: e~jyAtJQ9i)*9|)TNg><
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-08-14T15:29:57+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateO
rProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
|_ VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1


kali@kali: ~
Session Actions Edit View Help
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:1E:A0:ED (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Lin
ux; CPE: cpe:/o:linux:linux_kernel
Host script results:
|_clock-skew: mean: 59m59s, deviation: 2h00m00s, median: 0s
| smb-os-discovery:
| OS: Unix (Samba 3.0.20-Debian)
| Computer name: metasploitable
| NetBIOS computer name:
| Domain name: localdomain
| FQDN: metasploitable.localdomain
|_ System time: 2026-08-14T11:29:43-04:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown>
(unknown)
| smb-security-mode:
| account_used: guest
| authentication_level: user
| challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb2-time: Protocol negotiation failed (SMB2)
TRACEROUTE
HOP RTT ADDRESS
1 0.81 ms 192.168.56.102



nmap -n -p 21,22,80,443 192.168.56.102 helps me to scan the main specified ports. I can scan specific common services (FTP, SSH, HTTP). It tells only 4 ports state , if open or not.